

Basics 1: Units, Dimensions, and Normalization

I - Mathematical and computational foundations

A practical language for checking equations and constructing dimensionless plasma models

Textbook draft, version 1.0. Units specify a measurement convention, dimensions specify the physical type of a quantity, and normalization exposes the dimensionless parameters that organize a model. This chapter is designed for a reader with no prior plasma-physics training. It distinguishes exact identities from physical approximations and connects the central equations to later simulation modules.

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1 Learning objectives

After completing this note, the reader should be able to:

- Distinguish dimensions from units and convert among common plasma-physics units.
- Check dimensional homogeneity of algebraic, differential, and integral equations.
- Choose characteristic scales for length, time, velocity, density, pressure, magnetic field, and frequency.
- Nondimensionalize conservation laws and identify the resulting control parameters.
- Explain how normalization affects numerical conditioning and physics-informed loss functions.
- Document units and normalization at every software interface.

2 Prerequisites and reading strategy

- Elementary algebra, scientific notation, and differentiation.

On a first reading, emphasize physical meaning and dimensional checks. On a second reading, reproduce the derivations. Attempt the computational laboratory only after the limiting cases are understood.

3 Physical quantities, base dimensions, and SI units

A physical quantity is a numerical value multiplied by a unit. The same magnetic field may be written as

$$B = 2.5 \text{ T} = 2.5 \times 10^4 \text{ G}, \quad (3.1)$$

but its physical dimension is unchanged. In SI, the base dimensions needed here are mass M , length L , time T , electric current I , and thermodynamic temperature Θ . We write

$$[Q] = M^a L^b T^c I^d \Theta^e \quad (3.2)$$

to classify a quantity Q . Dimensions are independent of the human choice of meter, centimeter, second, or microsecond.

Examples are

$$[\rho] = ML^{-3}, \quad [\rho] = ML^{-1}T^{-2}, \quad (3.3)$$

$$[\mathbf{u}] = LT^{-1}, \quad [\mathbf{B}] = MT^{-2}I^{-1}, \quad (3.4)$$

$$[\mathbf{E}] = MLT^{-3}I^{-1}, \quad [\mu_0] = MLT^{-2}I^{-2}. \quad (3.5)$$

The magnetic-energy density $B^2/(2\mu_0)$ has dimension $ML^{-1}T^{-2}$, which is exactly the dimension of pressure. That equality is the dimensional reason magnetic pressure can enter the MHD momentum equation beside gas pressure.

4 Dimensional homogeneity as an error detector

Every term that is added to another term must have the same dimension. Consider

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \frac{1}{\mu_0} (\nabla \times \mathbf{B}) \times \mathbf{B}. \quad (4.1)$$

The inertial term scales as $\rho U^2/L$, the pressure force as p/L , and the magnetic force as $B^2/(\mu_0 L)$. All are force per volume. If a derivation accidentally uses $B/\mu_0 L$ rather than $B^2/\mu_0 L$, dimensional analysis rejects it immediately.

Dimensional consistency is necessary but not sufficient. The expression mv^2 and $2mv^2$ have the same dimension, but only one coefficient may be correct. Dimensional analysis also cannot detect a wrong sign or an omitted dimensionless factor such as 2π . It should therefore be the first verification step, not the last.

5 Characteristic scales

A characteristic scale is a representative magnitude chosen to organize a problem. Introduce

$$\mathbf{x} = L_0 \hat{\mathbf{x}}, \quad t = t_0 \hat{t}, \quad \mathbf{u} = U_0 \hat{\mathbf{u}}, \quad \rho = \rho_0 \hat{\rho}, \quad (5.1)$$

and similarly

$$\mathbf{B} = B_0 \hat{\mathbf{B}}, \quad p = p_0 \hat{p}, \quad \omega = \omega_0 \hat{\omega}. \quad (5.2)$$

For magnetized-fluid dynamics, natural choices include

$$V_{A0} = \frac{B_0}{\sqrt{\mu_0 \rho_0}}, \quad t_A = \frac{L_0}{V_{A0}}, \quad p_{B0} = \frac{B_0^2}{\mu_0}. \quad (5.3)$$

An orbit code may instead use a gyroperiod and gyroradius. A transport code may use a confinement time and minor radius. There is no universally correct normalization; there is only a normalization

that is explicit, internally consistent, and appropriate to the dominant scales.

6 Nondimensionalizing representative equations

The continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (6.1)$$

becomes

$$\frac{\rho_0}{t_0} \frac{\partial \hat{\rho}}{\partial \hat{t}} + \frac{\rho_0 U_0}{L_0} \hat{\nabla} \cdot (\hat{\rho} \hat{\mathbf{u}}) = 0. \quad (6.2)$$

Choosing $t_0 = L_0/U_0$ removes the prefactors. For magnetic induction with magnetic diffusivity η_m ,

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{u} \times \mathbf{B}) + \eta_m \nabla^2 \mathbf{B}, \quad (6.3)$$

the normalized equation is

$$\frac{\partial \hat{\mathbf{B}}}{\partial \hat{t}} = \hat{\nabla} \times (\hat{\mathbf{u}} \times \hat{\mathbf{B}}) + \frac{1}{R_m} \hat{\nabla}^2 \hat{\mathbf{B}}, \quad R_m = \frac{U_0 L_0}{\eta_m}. \quad (6.4)$$

The magnetic Reynolds number compares field advection with magnetic diffusion. In the momentum equation, the coefficient

$$\frac{B_0^2}{\mu_0 \rho_0 U_0^2} = \frac{1}{M_A^2} \quad (6.5)$$

introduces the Alfvén Mach number $M_A = U_0/V_A$.

7 Frequency, density, and temperature conventions

Angular frequency and cyclic frequency satisfy

$$\omega = 2\pi f. \quad (7.1)$$

A value in radians per second is not numerically equal to the value in hertz. Number density and mass density are related by

$$\rho = \sum_s m_s n_s. \quad (7.2)$$

For an Alfvén speed, using electron density alone would omit the dominant ion inertia.

The electron-volt is an energy unit:

$$1 \text{ eV} = 1.602176634 \times 10^{-19} \text{ J}. \quad (7.3)$$

Plasma temperature is often reported as $k_B T$ in electron-volts. A field named “temperature_eV”

normally stores an energy per particle, not kelvin. The code and the notes must state the convention.

8 Normalization in numerical simulation and PINNs

Floating-point calculations are more robust when important variables are of comparable magnitude. Normalized variables also make residual tolerances meaningful across modules.

In a PINN, suppose

$$\mathcal{L} = \mathcal{L}_{\text{mom}} + \mathcal{L}_{\nabla \cdot B}. \quad (8.1)$$

If dimensional residuals differ by many orders of magnitude, optimization may effectively ignore one equation. Dimensionless residuals make the relative weights interpretable, although adaptive weighting may still be needed.

A reproducible data product states the physical unit, normalization scale, sign convention, and whether the stored value is dimensional or normalized.

9 Computational laboratory

Create matching Python and MATLAB scripts that read B_0 , ion number density, ion mass, and major radius from JSON. Compute mass density, Alfvén speed, Alfvén time, angular frequency, and cyclic frequency. Double B_0 and quadruple ρ_0 and verify the expected scalings. Add tests that reject negative density and ambiguous frequency metadata.

10 Summary

- Dimensions classify physical quantities independently of the chosen unit system.
- Dimensional homogeneity is a powerful necessary check on every equation.
- Characteristic scales expose the dimensionless parameters controlling a model.
- Frequency, density, and temperature conventions must be explicit.
- Normalization affects numerical conditioning and machine-learning loss balance.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
L_0, t_0, U_0	reference scales	Characteristic length, time, and velocity.
V_A	Alfvén speed	$B/\sqrt{\mu_0\rho}$.
R_m	magnetic Reynolds number	Magnetic advection-to-diffusion ratio.
M_A	Alfvén Mach number	U/V_A .
$\hat{Q}$	normalized variable	Q/Q_0 .

11 Exercises

1. Derive the dimensions of ϵ_0 , electrical resistivity, current density, and electric potential.
2. Nondimensionalize $\partial_t u = D\partial_x^2 u$ and derive the diffusion time.
3. Show that $B^2/(\mu_0\rho)$ has dimensions of velocity squared.
4. Convert $\omega = 1.8 \times 10^6 \text{ s}^{-1}$ to hertz.
5. Construct a normalization table for the current Module 4 benchmark.
6. Explain how poor scaling can make a PINN satisfy one equation while violating another.

12 References

References

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